

Why might AI adversely affect exposed entry-level workers more than other age groups? One possibility is that AI disproportionately substitutes for workers using codified knowledge, including both the “book-learning” that forms the core of formal education and the insights in digital company data that can be codified by AI. AI may be less capable of replacing tacit knowledge, the idiosyncratic tips and tricks that accumulate with experience but which are never digitized.⁴ As young workers supply relatively more codified knowledge than tacit knowledge, they may face greater task replacement from AI in exposed occupations, leading to greater employment reallocation (Acemoglu and Autor, 2011). Older workers with accumulated tacit knowledge may face less task replacement. These benefits of uncoded knowledge may accrue less to non-college workers in occupations with low returns to experience. In other words, AI may be automating the *codifiable*, *checkable* tasks that historically justified entry-level headcount, while complementing the judgment-, client-, and process-intensive tasks performed by experienced workers.

Other explanations may contribute. AI may raise the leverage of experienced staff, increasing their effective span of control (Ide, 2025). Reduced hiring may also be the lowest-friction adjustment margin, compounded by inefficient incentives to train entry-level workers who may move firms (Becker, 1994; Garicano, 2025; Garicano and Rayo, 2025). Consequently firms may primarily shrink junior inflows rather than displace incumbents.

This combination of task substitution at the apprentice margin, complementarity at the expert margin, and quantity (not price) adjustment under wage ladders can explain why early-career employment falls while employment of older workers continues to grow. An important direction for research is to further model and test these predictions.

2 Related Literature

A growing body of research has sought to measure the employment effects of AI.⁵ This includes influential papers that established methodologies for estimating which occupations and tasks were susceptible to automation (Frey and Osborne, 2017; Brynjolfsson and Mitchell, 2017; Brynjolfsson et al., 2018; Felten et al., 2018, 2019; Webb, 2019; Felten et al., 2021). More recently, work such

⁴Ironically, one of the practical skills more likely to be learned on the job than in university computer science classes may be how to use AI for software development.

⁵There has also been extensive media discussion of this topic including Thompson (2025); Raman (2025); Roose (2025); The Economist (2025); Frick (2025). See Online Appendix B for further references and discussion.

as Eloundou et al. (2024); Felten et al. (2023); Gmyrek et al. (2023); Handa et al. (2025), and Tomlinson et al. (2025) adapted this approach for Generative AI, forming the basis for exposure metrics used in this analysis. While these studies identify potential disruption, our paper connects these exposure measures to actual employment changes.

Our work broadens insights from studies that find significant effects in more specific settings, such as on online freelance platforms (Hui et al., 2023; Demirci et al., 2025) or within individual firms (Brynjolfsson et al., 2025; Dillon et al., 2025).⁶ We measure labor market changes across occupations spanning the US economy.

In this sense our work complements a small but growing list of papers using economy-wide data to measure AI’s impact. Recent findings have been varied. Jiang et al. (2025) find AI exposure correlates with longer work hours in the U.S.⁷ Hampole et al. (2025) use job postings and LinkedIn profiles from Revelio labs from 2011 to 2023 to find limited employment impacts overall, with growing labor demand at firms offsetting relative declines in demand for exposed occupations. Chandar (2025b) uses data from the CPS to compare employment changes in more and less AI-exposed professions, finding little differential trend overall but noting difficulty with measuring changes for young workers given limited effective sample size.⁸ Johnston and Makridis (2025) find employment increases in state-industry pairs more exposed to AI using Quarterly Census of Employment and Wages (QCEW) data.⁹ These prior papers use data lacking either sufficient granularity or immediacy to reliably study employment changes by AI exposure and age (O’Brien, 2025).¹⁰ In contrast, this paper uses large-scale, close to real-time data to take a step towards resolving the ongoing debate on the employment effects of AI on young workers.

After we first publicly shared our results, work by Hosseini and Lichtinger (2025) and Klein Teeselink (2025) similarly found declines in AI-exposed entry-level employment using LinkedIn data from the

⁶See also Noy and Zhang (2023); Peng et al. (2023); Dell’Acqua et al. (2023).

⁷See also Acemoglu et al. (2022); Bonney et al. (2024); Bick et al. (2024); Hartley et al. (2025); Frank et al. (2025); Chen et al. (2025).

⁸See Dominski and Lee (2025); Gimbel et al. (2025); Eckhardt and Goldschlag (2025), which also use CPS data.

⁹Johnston and Makridis (2025) measure state-industry exposure by taking an average of Eloundou et al. (2024)’s occupational exposure weighted by state-industry employment. Industry-level labor market changes may be distinct from the occupation-level changes studied in this paper if firms make capital investments or become more productive in ways that increases overall labor demand (Hampole et al., 2025).

¹⁰As a comparison, the CPS surveyed between 44,000 and 51,000 employed individuals in total across all age groups in each month since 2021. Between 10,000 and 12,000 of these observations were in the outgoing rotation group and included earnings records. The data in our main analysis sample includes between 250,000 and 350,000 employed individuals in each month *just between the ages of 22 and 25*, all with earnings records.

US and UK. In contrast, the most recent version of Humlum and Vestergaard (2025) found minimal effects on entry-level earnings or hours worked in Denmark.¹¹

3 Data Description

3.1 Payroll Data

This study uses data from ADP, the largest payroll processing firm in America. The company provides payroll services for firms employing over 25 million workers in the US. We use this to track employment changes for workers in occupations measured as more or less exposed to artificial intelligence.

We make several restrictions for our main analysis sample. We include only workers with positive earnings, exclude part-time employees, and subset to people under age 70.¹²

The set of firms using payroll services changes over time as companies join or leave ADP’s platform. We maintain a consistent set of firms by keeping only companies that have employee earnings records for each month from January 2021 through September 2025.

ADP observes job titles for about 70% of workers in its system. We exclude workers without a recorded job title. There are over 7,000 standardized job titles, including “Search engineer optimization specialist,” “Enterprise content management manager,” and “Plant documentation control specialist.” The company’s internal research team maps each of these job titles to a 2010 Standard Occupational Classification (SOC) code, additionally using information such as the job description, industry, location, and other relevant data. We use these estimated SOC codes to merge our data to occupational AI exposure measures.

After these restrictions we have records on 3.5 and 5 million workers each month for our main analysis sample, though we consider robustness to alternative analyses such as allowing for firms to enter and leave the sample.

While the ADP data include millions of workers in each month, the distribution of firms using

¹¹An important direction for future work is to reconcile differences among these recent studies, assessing the extent they arise from differences in labor market institutions, measurement of AI adoption, statistical methodology, or other factors.

¹²While we observe the year of birth for each worker, for privacy reasons we do not observe the exact date of birth. We impute month of birth from the distribution of birth months in the United States using data from the Center for Disease Control and Prevention.

ADP services does not exactly match the distribution of firms across the broader US economy. Further details on differences in firm composition can be found in [Cajner et al. \(2018\)](#) and [ADP Research \(2025\)](#).¹³

3.2 Occupational AI Exposure

We use two approaches for measuring occupational exposure to AI. The first uses exposure measures from [Eloundou et al. \(2024\)](#), who estimate AI exposure by O*NET task using ChatGPT validated with human labeling. They construct occupational exposure measures by aggregating the task data to the 2018 SOC code level. We focus on the GPT-4 based β exposure measures from their paper.

The second approach we take uses generative AI usage data from the Anthropic Economic Index ([Handa et al., 2025](#)). This index reports the estimated share of queries pertaining to each O*NET task based on several million conversations with Claude, Anthropic’s generative AI model. It aggregates the data to the occupational level based on these task shares. One feature of the Anthropic Economic Index is that for each task it reports estimates of the share of queries pertaining to that task that are “automotive,” “augmentative,” or none of the above. We use this as an estimate of whether AI usage for an occupation is primarily complementary or substitutable with labor.¹⁴

We use a 2010 SOC code to 2018 SOC code crosswalk from the BLS to merge the exposure measures to the payroll data. Table [A1](#) shows example occupations for each AI exposure measure.

3.3 Other Data

To compare employment changes for teleworkable versus non-teleworkable occupations, we use data from [Dingel and Neiman \(2020\)](#). We use the Personal Consumption Expenditure index from the BLS to compute real earnings, indexed to October 2017. We use monthly Current Population Survey (CPS) data as a comparison for our main findings.

¹³[Cajner et al. \(2018\)](#) find a somewhat higher share of manufacturing and services firms compared to the QCEW using data from March 2016. They also find that ADP somewhat overrepresents firms in the Northeast. In addition, firms using ADP tend to grow faster on average than the typical firm in the US economy.

¹⁴[Handa et al. \(2025\)](#) use Claude to classify conversations into six categories. Directive and Feedback Loop conversations are considered Automotive; Task Iteration, Learning, and Validation are considered Augmentative. They also instruct the model to choose None of the above “liberally.” Table [A2](#) reproduces Table 1 from [Handa et al. \(2025\)](#) and shows more details about the automation and augmentation measures.